

2.1: Linear Functions

Definition. (Slope of a Nonvertical Line)

If (x_1, y_1) and (x_2, y_2) are any two distinct points on a nonvertical line L , then the slope m of L is given by

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

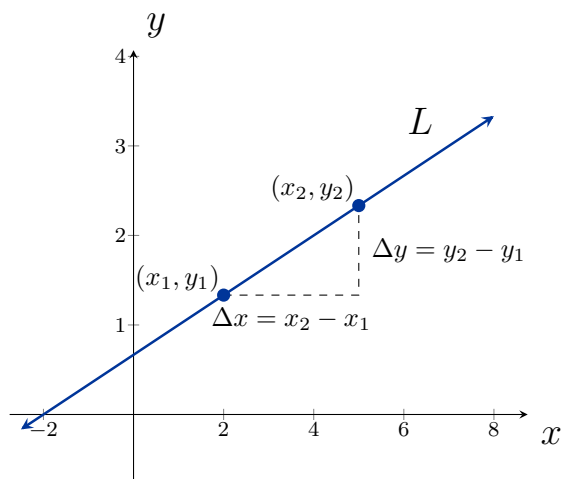
Example. Compute the slope of the line passing through the points

$$(x_1, y_1) = (1, 1) \text{ and } (x_2, y_2) = (4, 2)$$

$$(x_1, y_1) = (3, -1) \text{ and } (x_2, y_2) = (-2, 8)$$

$$(x_1, y_1) = (3, 2) \text{ and } (x_2, y_2) = (-1, 2)$$

$$(x_1, y_1) = (4, 1) \text{ and } (x_2, y_2) = (4, 4)$$



Definition. (Point-Slope Form of an Equation of a Line)

An equation of the line that has slope m and passes through the point (x_1, y_1) is given by

$$y - y_1 = m(x - x_1)$$

Example. Find the equation of the line going through the points

$$(x_1, y_1) = (-2, 1) \text{ and } (x_2, y_2) = (3, -2)$$

$$(x_1, y_1) = (3, 4) \text{ and } (x_2, y_2) = (-1, 4)$$

$$(x_1, y_1) = (2, 0) \text{ and } (x_2, y_2) = (2, 1)$$

Definition. (Slope-Intercept Form of an Equation of a Line)

An equation of the line that has slope m and intersects the y -axis at the point $(0, b)$ is given by

$$y = mx + b$$

Example. Rewrite the equations in the previous example in slope-intercept form.

Example. Marcus currently has 200 songs in his music collection. Every month, he adds 15 new songs. Write a formula for the number of songs, N , in his collection as a function of time, t , the number of months. How many songs will he own in a year?

Forms of Linear Equations

General form: $Ax + By = C$

Point-slope form: $y - y_1 = m(x - x_1)$

Slope-intercept form: $y = mx + b$

Vertical line: $x = a$

Horizontal line: $y = b$